Dataset: The Kaggle dataset "Cerebral Stroke Prediction-Imbalanced Dataset." Each row is one person, with 43,400 rows and 12 columns: id, gender, age, hypertension, heart disease, ever married, work type, residence type, average glucose level, BMI, smoking status, and stroke. Stroke is what we are predicting. Only about 1.8% of the people in the dataset had a stroke, so the data is very imbalanced.

What we are doing: Can machine learning classify stroke outcomes from demographic, lifestyle, and health information? This is for educational purposes only. We are not doctors. This project is Exploratory Data Analysis (EDA): practicing looking at data, cleaning up missing values with appropriate tools like the median, and turning categories into numbers. Later, we will use precision and recall to check how well our models find strokes.

First, we loaded the data in Python without changing the original file, so it stays safe and intact. Then we split the dataset into 70% training and 30% testing. We used stratify so both groups keep the same 1.8% stroke rate.

Next, we cleaned the missing values. We filled missing BMI with the median, which is the middle value, and labeled missing smoking status as "Unknown." We chose the median because outliers, like very high BMI values, could skew the average.

Then we encoded the text categories into numbers. Yes/no columns only needed 0/1, so no further conversion was necessary. Columns with more than two choices, like work type and smoking status, needed one-hot encoding, which gives each category its own 0/1 column. For example, "Unknown" smoking status became its own column. We also normalized average glucose level with a log transform, using the training data only. Glucose had a long tail of high values, and the log transform changed the shape of the distribution so it is more balanced. Scaling changes the range, while normalization changes the shape.

For the final project, we will use this cleaned data to train models like Logistic Regression and Random Forest to predict stroke, and we will judge them mostly on recall so we don't miss real strokes.